TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	66.4 ± 3.75
Correlation Stability	94.1 ± 0.30
Distribution Consistency	80.4 ± 4.13
Entropy Preservation	87.3 ± 0.46
Mean Z Score Drift	0.0 ± 0.02
Categorical Fidelity	98.4 ± 0.66
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	97.6 ± 0.92
Novelty Score	0.9 ± 0.00

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Dataset Audit: breast_cancer_wisconsin_diagnostic.csv

1. Dataset Portrait

Attribute	Value
Rows	569
Columns	31
Numeric features	30
Categorical features	1
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
radius1	14.127	14.280	+0.153	1.1%	OK
texture1	19.290	19.338	+0.048	0.3%	OK
perimeter1	91.969	92.948	+0.979	1.1%	OK
area1	654.889	670.692	+15.803	2.4%	OK
smoothness1	0.096	0.097	+0.001	0.6%	OK
compactness1	0.104	0.108	+0.004	3.4%	OK
concavity1	0.089	0.094	+0.005	5.8%	OK
concave_points1	0.049	0.051	+0.002	4.8%	OK
symmetry1	0.181	0.184	+0.002	1.3%	OK
fractal_dimension1	0.063	0.063	+0.000	0.4%	OK
radius2	0.405	0.404	-0.001	0.2%	OK
texture2	1.217	1.177	-0.040	3.2%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	66.4 ± 3.75
Correlation Stability (*2)	94.1 ± 0.30
Distribution Consistency (*3)	80.4 ± 4.13
Entropy Preservation (*4)	87.3 ± 0.46
Mean Z Score Drift (*5)	0.0 ± 0.02
Categorical Fidelity (*6)	98.4 ± 0.66
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	97.6 ± 0.92
Novelty Score (*11)	0.9 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9508771929824562	0.9412280701754387	0.9561403508771936	0.005263157894736836	
F1	0.9452155271956949	0.9276813853546984	0.9513875041390071	0.006171976943311985	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9403508771929825	0.9184210526315791	0.9245614035087719	-0.01578947368421051	
F1	0.9317642363488181	0.9039813559608577	0.9160922061848469	-0.01567203016397130 5	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9394736842105263	0.9324561403508772	0.9359649122807017	-0.00350877192982457 23	
F1	0.9314283604116784	0.9201394499522688	0.9286063251295825	-0.00282203528209576 96	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9692982456140351	0.9543859649122807	0.9657894736842104	-0.00350877192982455	
F1	0.9657812227588884	0.9480098283457336	0.9620362707764565	-0.00374495198243191	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9333333333333333	0.9245614035087721	0.9298245614035089	-0.00350877192982456 1	
F1	0.9242527459470589	0.9055695596640112	0.9198900917944295	-0.00436265415262937 5	

Model: RF

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9578947368421054	0.9473684210526315	0.9675438596491228	0.00964912280701756	
F1	0.9529949000149248	0.9399958444914048	0.963798337696993	0.01080343768206798	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9429824561403507	0.9280701754385966	0.9412280701754387	-0.00175438596491227	
F1	0.934337828992506	0.9172567327395791	0.933782813252566	-0.00055501573993999	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9710526315789474	0.9473684210526315	0.9614035087719298	-0.00964912280701757	
F1	0.9676338620041433	0.939380333798924	0.9553060165712335	-0.01232784543290979	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9605263157894737	0.9535087719298246	0.968421052631579	0.00789473684210525	
F1	0.9557853896138416	0.9476388725127667	0.9649256881917067	0.00914029857786515	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9473684210526315	0.9412280701754385	0.9298245614035089	-0.01754385964912280	
F1	0.938240784782178	0.9259199362221043	0.9200672649163032	-0.01817351986587487	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.99	10.38	122.8	1001.0	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871
2	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
3	19.69	21.25	130.0	1203.0	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999
4	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
5	20.29	14.34	135.1	1297.0	0.1003	0.1328	0.198	0.1043	0.1809	0.05883
6	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613
7	18.25	19.98	119.6	1040.0	0.09463	0.109	0.1127	0.074	0.1794	0.05742
8	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451
9	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
10	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243
11	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697
12	15.78	17.89	103.6	781.0	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082
13	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
14	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338
15	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682
16	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077
17	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922
18	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356
19	19.81	22.15	130.0	1260.0	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395
20	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766

B.1 Audit Trace: Fold 1 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	10.26	12.22	66.52	324.9	0.09594	0.08711	0.04334	0.01571	0.2037	0.06503
2	14.92	19.76	94.89	680.9	0.09587	0.1305	0.08487	0.04812	0.1505	0.06149
3	12.0	15.65	76.14	446.0	0.08261	0.06679	0.03112	0.01638	0.1689	0.06057
4	14.99	22.11	102.1	686.9	0.08473	0.07885	0.05862	0.0378	0.193	0.05746
5	12.76	13.78	81.78	499.0	0.08794	0.0763	0.02916	0.02402	0.1809	0.06183
6	18.25	18.58	120.2	1024.0	0.1099	0.1572	0.1657	0.09183	0.1538	0.06325
7	13.24	20.13	86.87	542.9	0.08284	0.1223	0.101	0.02833	0.1601	0.06432
8	11.2	29.37	70.67	386.0	0.07449	0.03558	0.0	0.0	0.106	0.05502
9	16.74	21.59	110.1	869.5	0.0961	0.1336	0.1348	0.06018	0.1896	0.05656
10	13.17	18.66	85.98	534.6	0.1158	0.1231	0.1226	0.0734	0.2128	0.06777

B.2 Audit Trace: Fold 2 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.87	18.68	89.77	584.1	0.08743	0.06945	0.01857	0.0188	0.1464	0.06287
2	12.76	29.43	81.72	502.5	0.07903	0.0958	0.06859	0.02833	0.1448	0.05859
3	12.94	16.02	82.71	516.6	0.1135	0.1535	0.1544	0.08815	0.2087	0.07613
4	15.66	19.38	102.7	761.7	0.09057	0.07112	0.03299	0.02957	0.1411	0.05912
5	14.44	24.99	96.39	641.2	0.08837	0.133	0.05892	0.05271	0.1767	0.06433
6	11.66	18.33	75.21	420.5	0.09774	0.1147	0.05263	0.03783	0.1598	0.06072
7	13.28	20.28	87.32	545.2	0.1041	0.1436	0.09847	0.06158	0.1974	0.06782
8	20.59	21.24	137.8	1320.0	0.1085	0.1644	0.2188	0.1121	0.1848	0.06222
9	12.18	20.52	77.22	458.7	0.08013	0.04038	0.02383	0.0177	0.1739	0.05677
10	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697

B.3 Audit Trace: Fold 3 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.6	24.49	77.58	412.7	0.103	0.07773	0.06664	0.02738	0.136	0.0644
2	14.22	23.81	93.63	623.9	0.09463	0.2225	0.2733	0.0598	0.2521	0.06232
3	14.74	20.21	94.87	662.7	0.08757	0.1108	0.0905	0.03562	0.1618	0.0602
4	15.49	23.95	98.92	766.6	0.0784	0.06669	0.03649	0.02877	0.184	0.05697
5	13.66	15.79	87.76	571.1	0.0995	0.07624	0.03974	0.02995	0.1937	0.0571
6	14.54	21.53	96.73	642.7	0.1137	0.1697	0.1357	0.09029	0.235	0.07371
7	10.97	17.2	71.73	371.5	0.08915	0.1113	0.09457	0.03613	0.1489	0.0664
8	17.29	22.13	114.4	947.8	0.08999	0.1273	0.09697	0.07507	0.2108	0.05464
9	20.31	27.06	132.9	1288.0	0.1	0.1088	0.1519	0.09333	0.1814	0.05572
10	11.9	14.65	78.11	432.8	0.1152	0.1296	0.0371	0.03003	0.1995	0.07839

B.4 Audit Trace: Fold 4 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	18.25	19.98	118.7	1052.0	0.0943	0.1076	0.09938	0.06431	0.1739	0.05699
2	13.48	23.84	88.1	559.2	0.09258	0.1047	0.05375	0.03085	0.1727	0.0602
3	16.65	21.38	109.3	904.6	0.1119	0.1457	0.1491	0.08481	0.2027	0.0633
4	12.04	17.93	76.09	446.2	0.08372	0.04994	0.02109	0.02173	0.1739	0.06048
5	12.62	20.76	81.92	480.4	0.09667	0.1147	0.08175	0.02166	0.1761	0.06659
6	10.9	14.93	68.51	361.6	0.08792	0.04571	0.01993	0.01257	0.1897	0.06083
7	20.09	23.86	134.7	1247.0	0.108	0.1838	0.2283	0.128	0.2249	0.07469
8	11.52	18.75	73.34	409.0	0.09524	0.05473	0.03036	0.02278	0.192	0.05907
9	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
10	17.75	28.03	117.3	981.6	0.09997	0.1314	0.1698	0.08293	0.1713	0.05916

B.5 Audit Trace: Fold 5 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.7	13.27	87.76	575.5	0.09198	0.06221	0.04548	0.03263	0.162	0.05657
2	13.2	13.72	84.1	538.4	0.08363	0.04458	0.007756	0.00625	0.1583	0.05501
3	13.34	20.54	86.34	529.4	0.08445	0.08259	0.04072	0.02369	0.1809	0.05594
4	13.17	22.61	85.42	530.2	0.1131	0.1752	0.145	0.08543	0.2548	0.07016
5	14.25	22.55	92.87	629.9	0.1131	0.198	0.1697	0.08172	0.2188	0.07769
6	8.618	20.98	54.66	221.8	0.1082	0.08333	0.02031	0.009615	0.2538	0.07126
7	8.726	15.83	55.84	230.9	0.115	0.08201	0.04132	0.01924	0.1649	0.07633
8	19.4	18.18	127.2	1145.0	0.1037	0.1442	0.1626	0.09464	0.1893	0.05892
9	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697
10	12.19	13.29	79.08	455.8	0.1066	0.09509	0.02855	0.02882	0.188	0.06471

B.6 Audit Trace: Fold 6 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.11	24.04	85.63	530.2	0.1034	0.09769	0.1126	0.02854	0.1647	0.06677
2	10.08	15.11	64.41	313.1	0.1007	0.06889	0.02297	0.01952	0.1883	0.0644
3	20.09	19.66	131.2	1247.0	0.08355	0.08468	0.09042	0.06576	0.1893	0.05325
4	13.69	16.07	88.27	580.6	0.07944	0.06059	0.03393	0.02847	0.1714	0.05669
5	11.25	19.86	71.49	386.8	0.08511	0.1108	0.09252	0.02867	0.2031	0.06612
6	17.75	21.02	114.2	981.6	0.09847	0.1289	0.1204	0.07762	0.2116	0.0558
7	11.6	12.84	74.34	412.6	0.08983	0.07525	0.04196	0.0335	0.162	0.06582
8	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
9	15.46	11.89	102.5	736.9	0.1257	0.1555	0.2032	0.1097	0.1966	0.07069
10	14.9	22.53	102.1	685.0	0.09947	0.2225	0.2733	0.09711	0.2041	0.06898

B.7 Audit Trace: Fold 7 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.2	17.43	85.63	542.9	0.0806	0.03789	0.004473	0.006588	0.1667	0.05234
2	20.44	21.78	133.8	1264.0	0.09446	0.1022	0.09799	0.07488	0.1946	0.05581
3	13.61	20.83	87.88	575.5	0.1016	0.1154	0.1226	0.04105	0.1814	0.06419
4	14.81	14.7	94.21	671.4	0.09855	0.08501	0.03888	0.02315	0.1769	0.05549
5	9.042	13.9	58.79	245.2	0.09968	0.05272	0.0	0.0	0.183	0.07187
6	11.71	16.67	75.46	419.8	0.1128	0.09263	0.06895	0.02017	0.1828	0.06612
7	19.89	20.26	130.5	1214.0	0.1037	0.131	0.1411	0.09431	0.1802	0.06188
8	17.42	25.56	114.5	948.0	0.1006	0.1146	0.1682	0.06597	0.1308	0.05866
9	11.95	14.96	77.23	426.7	0.1158	0.1206	0.01171	0.01787	0.2459	0.06581
10	14.69	13.98	98.22	656.1	0.1031	0.1836	0.145	0.063	0.2086	0.07406

B.8 Audit Trace: Fold 8 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	9.731	12.44	61.24	290.9	0.08481	0.04695	0.01012	0.005495	0.1769	0.06412
2	12.75	16.7	82.51	492.1	0.1218	0.0721	0.04944	0.04531	0.1828	0.06168
3	12.34	24.04	81.15	462.9	0.1186	0.2087	0.1188	0.06526	0.2397	0.07799
4	15.22	22.41	98.92	731.3	0.0784	0.0629	0.04445	0.03263	0.177	0.05526
5	16.13	22.91	106.2	817.7	0.08206	0.0629	0.04209	0.02837	0.1664	0.05526
6	14.8	17.66	94.87	664.9	0.09686	0.08836	0.05192	0.03783	0.1594	0.05915
7	13.05	18.59	85.09	512.0	0.1082	0.1304	0.09603	0.05603	0.2035	0.06501
8	10.16	19.59	64.73	311.7	0.1003	0.07504	0.005025	0.01116	0.1791	0.06331
9	16.24	18.77	108.8	805.1	0.1066	0.1802	0.1948	0.09052	0.1876	0.06684
10	14.87	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.04951	0.1487	0.05748

B.9 Audit Trace: Fold 9 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.93	20.74	124.4	1001.0	0.1096	0.11	0.1508	0.08488	0.2124	0.0622
2	12.06	13.44	78.04	447.8	0.08605	0.06679	0.02643	0.02402	0.1583	0.06506
3	16.35	28.08	108.3	840.4	0.09831	0.1088	0.1007	0.05364	0.1769	0.05699
4	12.62	16.32	81.25	481.6	0.0876	0.07255	0.05485	0.01863	0.158	0.05935
5	12.16	18.9	78.07	458.7	0.08673	0.069	0.01994	0.01917	0.1667	0.06287
6	11.47	17.31	73.66	403.5	0.09076	0.07694	0.03265	0.01863	0.1388	0.06043
7	14.97	16.95	96.22	685.9	0.09855	0.07885	0.02602	0.03781	0.178	0.0565
8	11.43	15.39	73.06	399.8	0.09639	0.06889	0.03503	0.02875	0.1734	0.05865
9	11.81	17.39	75.27	428.9	0.1007	0.05562	0.02353	0.01553	0.1718	0.0578
10	10.57	18.32	66.82	340.9	0.08142	0.04462	0.01993	0.01111	0.2372	0.05768

B.10 Audit Trace: Fold 10 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.51	13.9	87.91	561.3	0.1059	0.1807	0.08448	0.04223	0.1668	0.0731
2	10.96	14.93	70.79	371.5	0.07274	0.05055	0.02475	0.01374	0.1869	0.05688
3	14.41	13.47	94.21	656.4	0.09906	0.1192	0.08007	0.06142	0.1662	0.06121
4	12.36	19.02	78.94	464.5	0.09231	0.166	0.06651	0.02978	0.1995	0.0731
5	18.22	18.7	120.2	1033.0	0.1008	0.1041	0.1044	0.07507	0.1794	0.05961
6	13.82	15.79	90.43	592.6	0.1007	0.07862	0.04944	0.04528	0.2075	0.062
7	13.71	18.68	88.73	571.0	0.09916	0.107	0.05385	0.03783	0.1714	0.06843
8	11.94	18.24	75.71	437.6	0.08261	0.04751	0.01972	0.01349	0.1868	0.0611
9	10.51	23.09	66.85	334.2	0.1015	0.06797	0.02495	0.01875	0.1695	0.06556
10	11.75	17.56	75.89	422.9	0.1073	0.09713	0.05282	0.0444	0.1598	0.06677